PART I

Reading 1 "Beowulf"

Historical Background

➔ The epic poem *Beowulf*, written in Old English, is the earliest existing Germanic epic and one of four surviving Anglo-Saxon manuscripts. Although *Beowulf* was written by an anonymous Englishman in Old English, the tale takes place in that part of Scandinavia from which Germanic tribes emigrated to England. Beowulf comes from Geatland, the southeastern part of what is now Sweden. Hrothgar, king of the Danes, lives near what is now Leire, on Zealand, Denmark's largest island. The *Beowulf* epic contains three major tales about Beowulf and several minor tales that reflect a rich Germanic oral tradition of myths, legends, and folklore.

➔ The *Beowulf* warriors have a foot in both the Bronze and Iron Ages. Their mead-halls reflect the wealthy living of the Bronze Age Northmen, and their wooden shields, wood-shafted spears, and bronze-hilted swords are those of the Bronze Age warrior. However, they carry iron-tipped spears, and their best swords have iron or iron-edged blades. Beowulf also orders an iron shield for his fight with a dragon. Iron replaced bronze because it produced a blade with a cutting edge that was stronger and sharper. The Northmen learned how to forge iron in about 500 B.C. Although they had been superior to the European Celts in bronze work, it was the Celts who taught them how to make and design iron work. Iron was accessible everywhere in Scandinavia, usually in the form of "bog-iron" found in the layers of peat in peat bogs.

The *Beowulf* epic also reveals interesting aspects of the lives of the Anglo-Saxons who lived in England at the time of the anonymous *Beowulf* poet. The Germanic tribes, including the Angles, the Saxons, and the Jutes, invaded England from about A.D. 450 to 600. By the time of the *Beowulf* poet, Anglo-Saxon society in England was neither primitive nor uncultured. [A]

➔ Although the *Beowulf* manuscript was written in about A.D. 1000, it was not discovered until the seventeenth century. [B] Scholars do not know whether *Beowulf* is the sole surviving epic from a flourishing Anglo-Saxon literary period that produced other great epics or whether it was unique even in its own time. [C] Many scholars think that the epic was probably written sometime between the late seventh century and the early ninth century. If they are correct, the original manuscript was probably lost during the ninth-century Viking invasions of Anglia, in which the Danes destroyed the Anglo-Saxon monasteries and their great libraries. However, other scholars think that the poet's favorable attitude toward the Danes must place the epic's composition after the Viking invasions and at the start of the eleventh century, when this *Beowulf* manuscript was written.

➔ The identity of the *Beowulf* poet is also uncertain. D He apparently was a Christian who loved the pagan heroic tradition of his ancestors and blended the values of the pagan hero with the Christian values of his own country and time. Because he wrote in the Anglian dialect, he probably was either a monk in a monastery or a poet in an Anglo-Saxon court located north of the Thames River.

Appeal and Value

Beowulf interests contemporary readers for many reasons. First, it is an outstanding adventure story. Grendel, Grendel's mother, and the dragon are marvelous characters, and each fight is unique, action-packed, and exciting. Second, Beowulf is a very appealing hero. He is the perfect warrior, combining extraordinary strength, skill, courage, and loyalty. Like Hercules, he devotes his life to making the world a safer place. He chooses to risk death in order to help other people, and he faces his inevitable death with heroism and dignity. Third, the *Beowulf* poet is interested in the psychological aspects of human behavior. For example, the Danish hero's welcoming speech illustrates his jealousy of Beowulf. The behavior of Beowulf's warriors in the dragon fight reveals their cowardice. Beowulf's attitudes toward heroism reflect his maturity and experience, while King Hrothgar's attitudes toward life show the experiences of an aged nobleman.

Finally, the *Beowulf* poet exhibits a mature appreciation of the transitory nature of human life and achievement. In Beowulf, as in the major epics of other cultures, the hero must create a meaningful life in a world that is often dangerous and uncaring. He must accept the inevitability of death. He chooses to reject despair; instead, he takes pride in himself and in his accomplishments, and he values human relationships.

1. According to paragraph 1, which of the following is true about *Beowulf*?

   Ⓐ It is the only manuscript from the Anglo-Saxon period.
   Ⓑ The original story was written in a German dialect.
   Ⓒ The author did not sign his name to the poem.
   Ⓓ It is one of several epics from the first century.

   Paragraph 1 is marked with an arrow [➔].

2. According to paragraph 4, why do many scholars believe that the original manuscript for *Beowulf* was lost?

   Ⓐ Because it is not like other manuscripts
   Ⓑ Because many libraries were burned
   Ⓒ Because the Danes were allies of the Anglo-Saxons
   Ⓓ Because no copies were found in monasteries

   Paragraph 4 is marked with an arrow [➔].

3. Why does the author of this passage use the word "apparently" in paragraph 5?

   Ⓐ He is not certain that the author of *Beowulf* was a Christian.
   Ⓑ He is mentioning facts that are obvious to the readers.
   Ⓒ He is giving an example from a historical reference.
   Ⓓ He is introducing evidence about the author of *Beowulf.*

   Paragraph 5 is marked with an arrow [➔].

4. Why does the author mention "bog-iron" in paragraph 2?

   Ⓐ To demonstrate the availability of iron in Scandinavia
   Ⓑ To prove that iron was better than bronze for weapons
   Ⓒ To argue that the Celts provided the materials to make iron
   Ⓓ To suggest that 500 B.C. was the date that the Iron Age began

   Paragraph 2 is marked with an arrow [➔].

5. The word <u>unique</u> in the passage is closest in meaning to

   Ⓐ old
   Ⓑ rare
   Ⓒ perfect
   Ⓓ weak

6. The word <u>exhibits</u> in the passage is closest in meaning to

   Ⓐ creates
   Ⓑ demonstrates
   Ⓒ assumes
   Ⓓ terminates

7. The word <u>reject</u> in the passage is closest in meaning to

   Ⓐ manage
   Ⓑ evaluate
   Ⓒ refuse
   Ⓓ confront

8. Which of the sentences below best expresses the information in the highlighted statement in the passage? The other choices change the meaning or leave out important information.

   Ⓐ Society in Anglo-Saxon England was both advanced and cultured.
   Ⓑ The society of the Anglo-Saxons was not primitive or cultured.
   Ⓒ The Anglo-Saxons had a society that was primitive, not cultured.
   Ⓓ England during the Anglo-Saxon society was advanced, not cultured.

9. Look at the four squares [■] that show where the following sentence could be inserted in the passage.

   Moreover, they disagree as to whether this *Beowulf* is a copy of an earlier manuscript.

   Where could the sentence best be added?

   Click on a square [■] to insert the sentence in the passage.

10. **Directions:** An introduction for a short summary of the passage appears below. Complete the summary by selecting the THREE answer choices that mention the most important points in the passage. Some sentences do not belong in the summary because they express ideas that are not included in the passage or are minor points from the passage. ***This question is worth 2 points.***

    ***Beowulf* is the oldest Anglo-Saxon epic poem that has survived to the present day.**

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Answer Choices

A The Northmen were adept in crafting tools and weapons made of bronze, but the Celts were superior in designing and working in iron.

B In the Viking invasions of England, the Danish armies destroyed monasteries, some of which contained extensive libraries.

C King Hrothgar and Beowulf become friends at the end of their lives, after having spent decades opposing each other on the battlefield.

D The poem chronicles life in Anglo-Saxon society during the Bronze and Iron Ages when Germanic tribes were invading England.

E Although *Beowulf* was written by an anonymous poet, probably a Christian, about 1000 A.D., it was not found until the seventeenth century.

F *Beowulf* is still interesting because it has engaging characters, an adventurous plot, and an appreciation for human behavior and relationships.

PART II

Reading 2 "Thermoregulation"

➔ Mammals and birds generally maintain body temperature within a narrow range (36–38°C for most mammals and 39–42°C for most birds) that is usually considerably warmer than the environment. Because heat always flows from a warm object to cooler surroundings, birds and mammals must counteract the constant heat loss. This maintenance of warm body temperature depends on several key adaptations. The most basic mechanism is the high metabolic rate of endothermy itself. Endotherms can produce large amounts of metabolic heat that replace the flow of heat to the environment, and they can vary heat production to match changing rates of heat loss. Heat production is increased by such muscle activity as moving or shivering. In some mammals, certain hormones can cause mitochondria to increase their metabolic activity and produce heat instead of ATP. This **nonshivering thermogenesis (NST)** takes place throughout the body, but some mammals also have a tissue called **brown fat** in the neck and between the shoulders that is specialized for rapid heat production. Through shivering and NST, mammals and birds in cold environments can increase their metabolic heat production by as much as 5 to 10 times above the minimal levels that occur in warm conditions.

➔ Another major thermoregulatory adaptation that evolved in mammals and birds is insulation (hair, feathers, and fat layers), which reduces the flow of heat and lowers the energy cost of keeping warm. Most land mammals and birds react to cold by raising their fur or feathers, thereby trapping a thicker layer of air. [A] Humans rely more on a layer of fat just beneath the skin as insulation; goose bumps are a vestige of hair-raising left over from our furry ancestors. [B] Vasodilation and vasoconstriction also regulate heat exchange and may contribute to regional temperature differences within the animal. [C] For example, heat loss from a human is reduced when arms and legs cool to several degrees below the temperature of the body core, where most vital organs are located. [D]

➔ Hair loses most of its insulating power when wet. Marine mammals such as whales and seals have a very thick layer of insulation fat called blubber, just under the skin. Marine mammals swim in water colder than their body core temperature, and many species spend at least part of the year in nearly freezing polar seas. The loss of heat to water occurs 50 to 100 times more rapidly than heat loss to air, and the skin temperature of a marine mammal is close to water temperature. Even so, the blubber insulation is so effective that marine mammals maintain body core temperatures of about 36–38°C with metabolic rates about the same as those of land mammals of similar size. The flippers or tail of a whale or seal lack insulating blubber, but countercurrent heat exchangers greatly reduce heat loss in these extremities, as they do in the legs of many birds.

➔ Through metabolic heat production, insulation, and vascular adjustments, birds and mammals are capable of astonishing feats of thermoregulation. For example, small birds called chickadees, which weigh only 20 grams, can remain active and hold body temperature nearly constant at 40°C in environmental temperatures as low as –40°C—as long as they have enough food to supply the large amount of energy necessary for heat production.

Many mammals and birds live in places where thermoregulation requires cooling off as well as warming. For example, when a marine mammal moves into warm seas, as many whales do when they reproduce, excess metabolic heat is removed by vasodilation of numerous blood vessels in the outer layer of the skin. In hot climates or when vigorous exercise adds large amounts of metabolic heat to the body, many terrestrial mammals and birds may allow body temperature to rise by several degrees, which enhances heat loss by increasing the temperature gradient between the body and a warm environment.

➔ Evaporative cooling often plays a key role in dissipating the body heat. If environmental temperature is above body temperature, animals gain heat from the environment as well as from metabolism, and evaporation is the only way to keep body temperature from rising rapidly. Panting is important in birds and many mammals. Some birds have a pouch richly supplied with blood vessels in the floor of the mouth; fluttering the pouch increases evaporation. Pigeons can use evaporative cooling to keep body temperature close to 40°C in air temperatures as high as 60°C, as long as they have sufficient water. Many terrestrial mammals have sweat glands controlled by the nervous system. Other mechanisms that promote evaporative cooling include spreading saliva on body surfaces, an adaptation of some kangaroos and rodents for combating severe heat stress. Some bats use both saliva and urine to enhance evaporative cooling.

Glossary
ATP: energy that drives certain reactions in cells
mitochondria: a membrane of ATP

11. According to paragraph 1, the most fundamental adaptation to maintain body temperature is

Ⓐ the heat generated by the metabolism
Ⓑ a shivering reflex in the muscles
Ⓒ migration to a warmer environment
Ⓓ higher caloric intake to match heat loss

Paragraph 1 is marked with an arrow [➔].

12. In paragraph 6, the author states that evaporative cooling is often accomplished by all of the following methods EXCEPT

Ⓐ by spreading saliva over the area
Ⓑ by urinating on the body
Ⓒ by panting or fluttering a pouch
Ⓓ by immersing themselves in water

Paragraph 6 is marked with an arrow [➔].

13. Based on information in paragraph 1, which of the following best explains the term "thermogenesis"?

Ⓐ Heat loss that must be reversed
Ⓑ The adaptation of brown fat tissue in the neck
Ⓒ The maintenance of healthy environmental conditions
Ⓓ Conditions that affect the metabolism

Paragraph 1 is marked with an arrow [➔].

14. Why does the author mention chickadees in paragraph 4?

Ⓐ To discuss an animal that regulates heat very well
Ⓑ To demonstrate why chickadees have to eat so much
Ⓒ To mention an exception to the rules of thermoregulation
Ⓓ To give a reason for heat production in small animals

Paragraph 4 is marked with an arrow [➔].

15. The word <u>minimal</u> in the passage is closest in meaning to

Ⓐ most recent
Ⓑ most active
Ⓒ newest
Ⓓ smallest

16. The word <u>regulate</u> in the passage is closest in meaning to

Ⓐ protect
Ⓑ create
Ⓒ reduce
Ⓓ control

17. The word enhance in the passage is closest in meaning to

Ⓐ simplify
Ⓑ improve
Ⓒ replace
Ⓓ interrupt

18. Which of the sentences below best expresses the information in the highlighted statement in the passage? The other choices change the meaning or leave out important information.

Ⓐ An increase in heat production causes muscle activity such as moving or shivering.
Ⓑ Muscle activity like moving and shivering will increase heat production.
Ⓒ Moving and shivering are muscle activities that increase with heat.
Ⓓ When heat increases, the production of muscle activity also increases.

19. Look at the four squares [■] that show where the following sentence could be inserted in the passage.

The insulating power of a layer of fur or feathers mainly depends on how much still air the layer traps.

Where could the sentence best be added?

Click on a square [■] to insert the sentence in the passage.

20. **Directions:** An introduction for a short summary of the passage appears below. Complete the summary by selecting the THREE answer choices that mention the most important points in the passage. Some sentences do not belong in the summary because they express ideas that are not included in the passage or are minor points from the passage. ***This question is worth 2 points.***

Thermoregulation is the process by which animals control body temperatures within healthy limits.

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Answer Choices

A Although hair can be a very efficient insulation when it is dry and it can be raised, hair becomes ineffective when it is submerged in cold water.

B Some animals with few adaptations for thermoregulation migrate to moderate climates to avoid the extreme weather in the polar regions and the tropics.

C Mammals and birds use insulation to mitigate heat loss, including hair and feathers that can be raised to trap air as well as fat or blubber under the skin.

D Some birds have a special pouch in the mouth, which can be fluttered to increase evaporation and decrease their body temperatures by as much as 20°C.

E Endotherms generate heat by increasing muscle activity, by releasing hormones into their blood streams, or by producing heat in brown fat tissues.

F Panting, sweating, and spreading saliva or urine on their bodies are all options for the evaporative cooling of animals in hot environmental conditions.